Statistical Methods & Machine Learning 1

Class #10

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November 5, 2024

Outline

- Review
  - Regression assumptions
  - Unusual observations
  - Multiple testing

Regression assumptions

OLS Assumptions

Will see these assumptions presented a variety of ways (e.g., some are often combined), but they boil down to the following:

1. There is a linear relationship between the predictor(s) X and the outcome Y. – Relatively simple assumption to relax as we will see.
2. Variable, $X_1, \ldots, X_n$, are fixed constants such that $S_{XX} > 0$
3. Variable, $\epsilon_1, \ldots, \epsilon_n$, are random variables
4. Zero Error Mean: $E(\epsilon_i) = 0$ for $i = 1, \ldots, n$
5. Uncorrelated errors: $\epsilon_1, \epsilon_2, \ldots, \epsilon_n$ are not correlated – i.e., knowing value of ϵ_1 tells you nothing about $\epsilon_2, \ldots, \epsilon_n$
6. Constant error variance: $\mathrm{V}(\epsilon_i) = \sigma^2$ -- AKA homoscedasticity
7. Error normality: $\epsilon_i \sim N(0, \sigma^2)\ for\ i = 1, \ldots, n$

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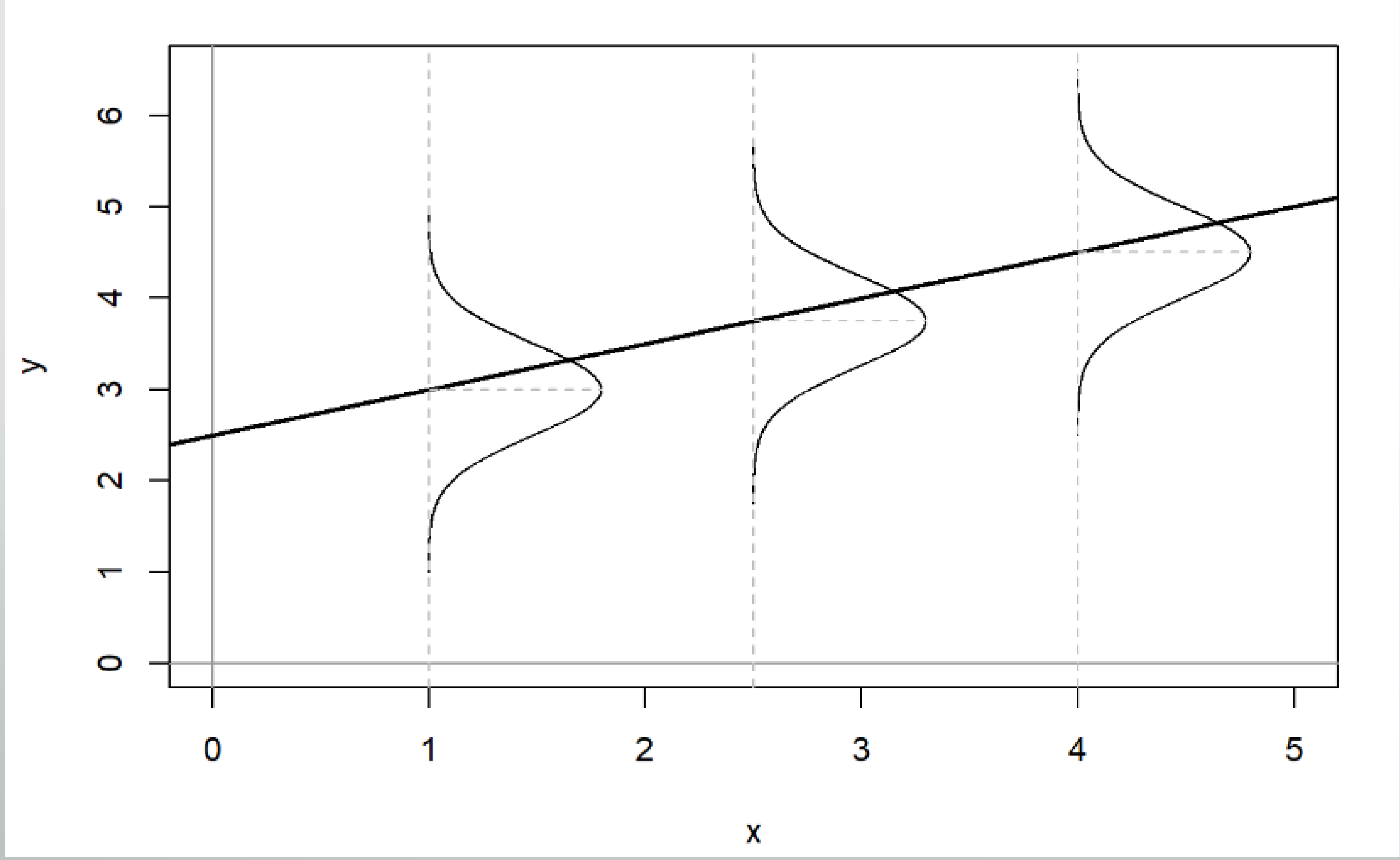
OLS Assumptions

Pay particular attention to assumptions 4-7

- Taken together they state that our errors are assumed to be normal, independent, and identically distributed (Normal I.I.D).



Jenkins-Smith, Ripberger, Copeland, et. al.

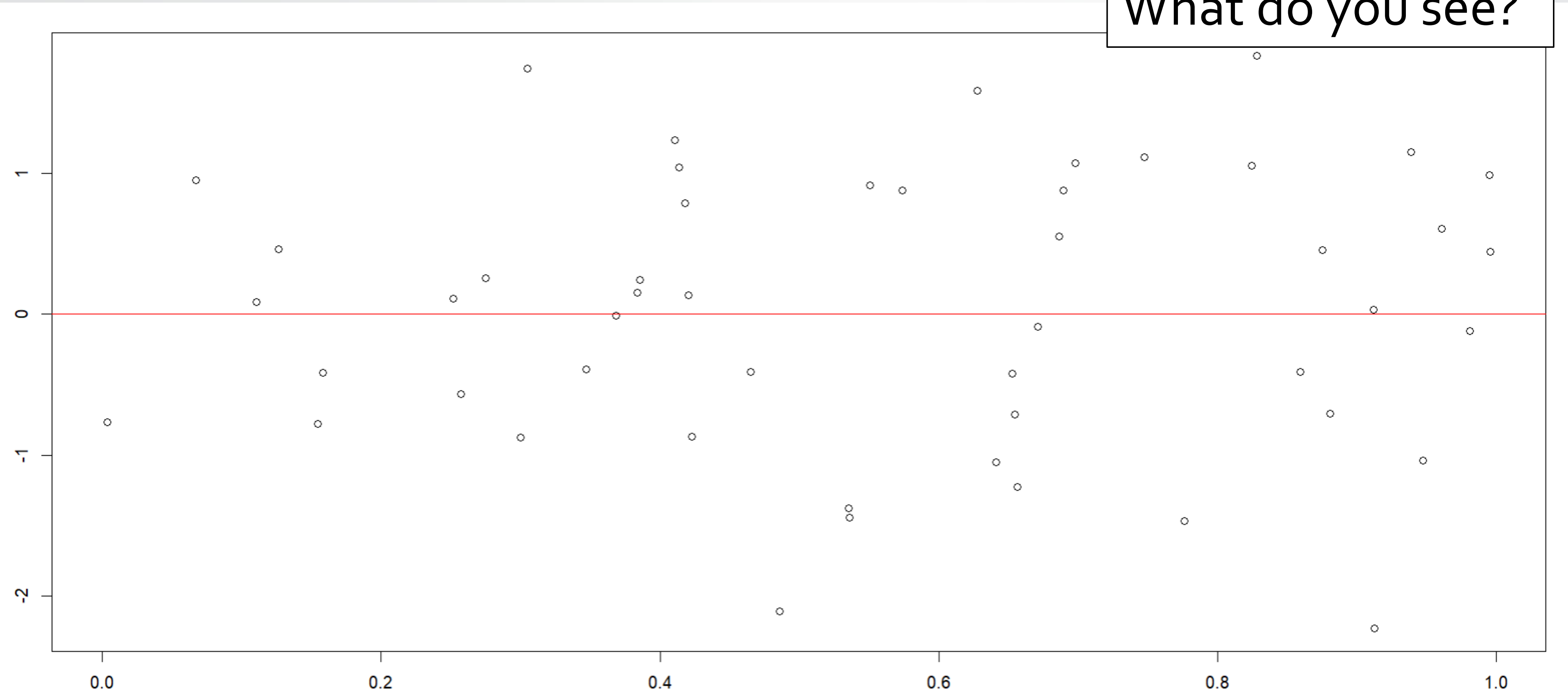
Impact of assumption violations

- Dependence of errors can cause problems as it means there is less information in your dataset than sample size would lead you to believe.
  - Think your estimates are more precise than they are.
- Non-constant error variance can cause problems with inaccurate inferences due to poorly quantified precision.
  - Prediction uncertainty particularly susceptible.
- Non-normal errors can also lead to inaccurate inferences, but for larger datasets inferences will be more robust to moderate violations due to the central limit theorem.
- Violation of zero mean error assumption can indicate issues with error dependence, error normality, and/or non constant error variance.

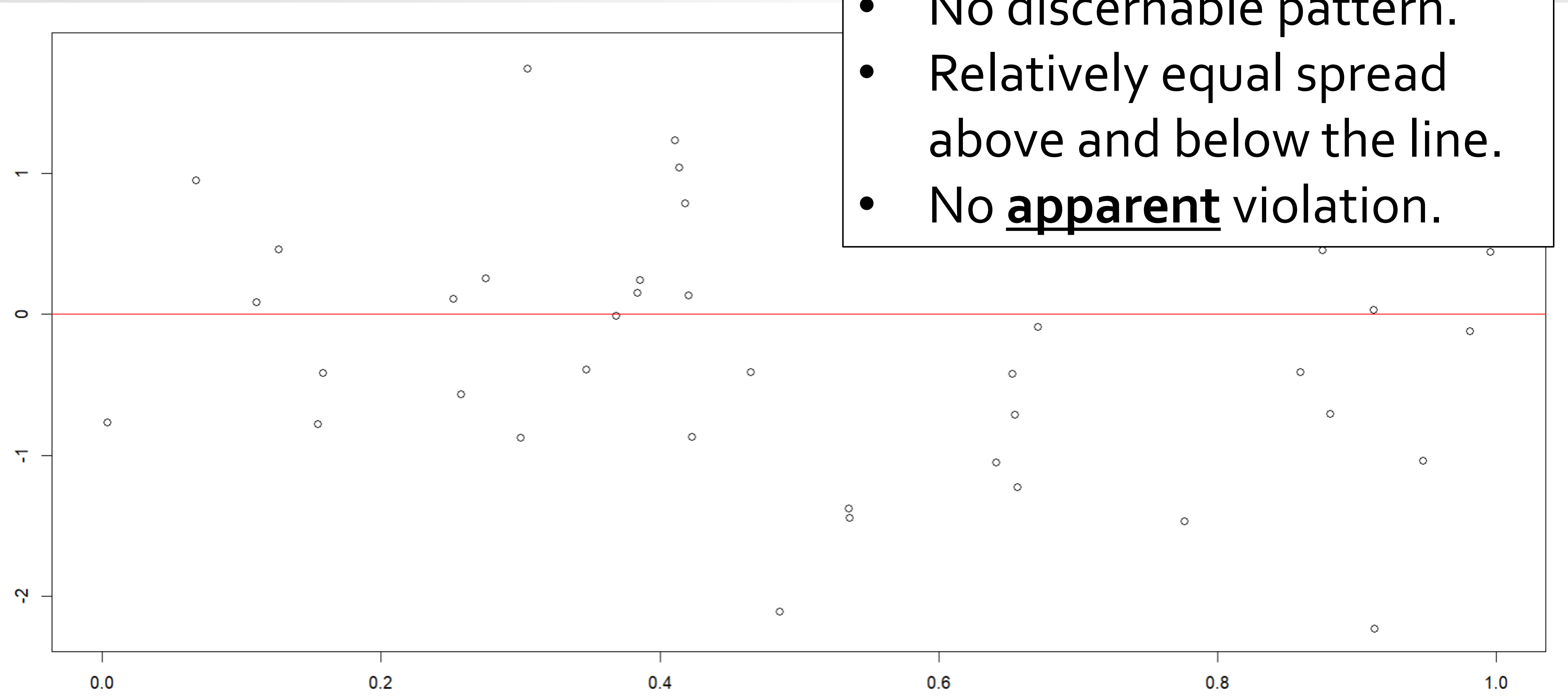
Zero mean error + homoscedasticity

- Let’s look at a few different scenarios to see what these violations might look like graphically.
- To do this we will fit a scatter plot with our fitted values for a given model on the x axis and the residuals from the same model on y axis

Zero mean error + homoscedasticity

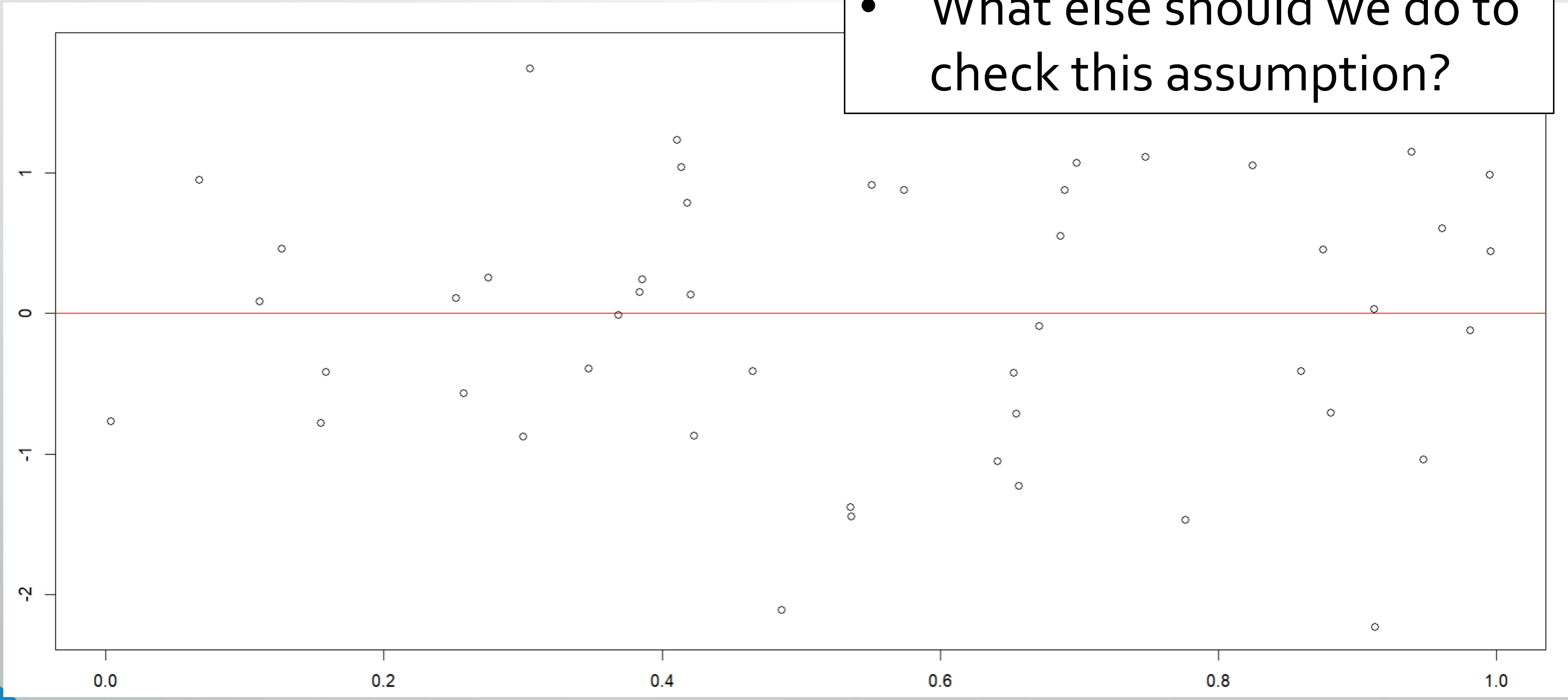


Zero mean error + homoscedasticity



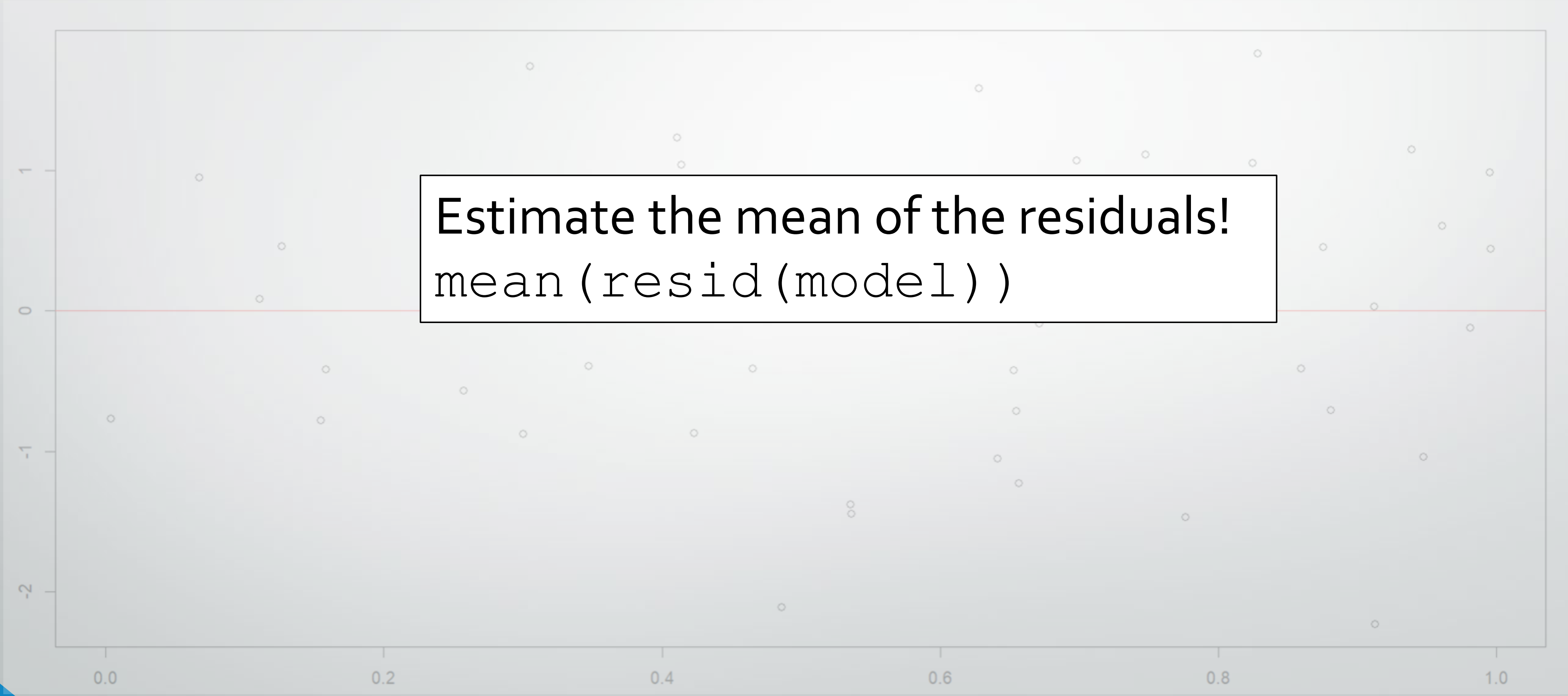
- No discernable pattern.
- Relatively equal spread above and below the line.
- No **apparent** violation.

Zero mean error



- What else should we do to check this assumption?

Zero mean error

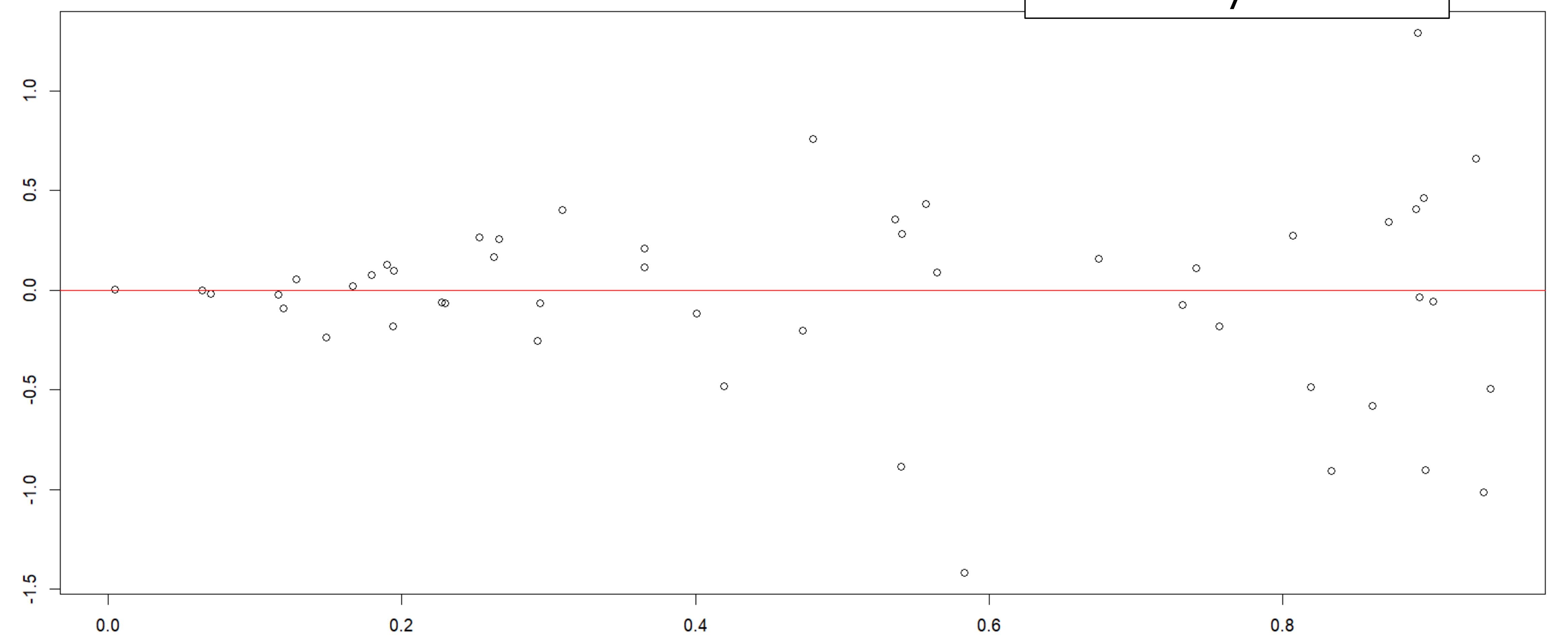


Estimate the mean of the residuals!

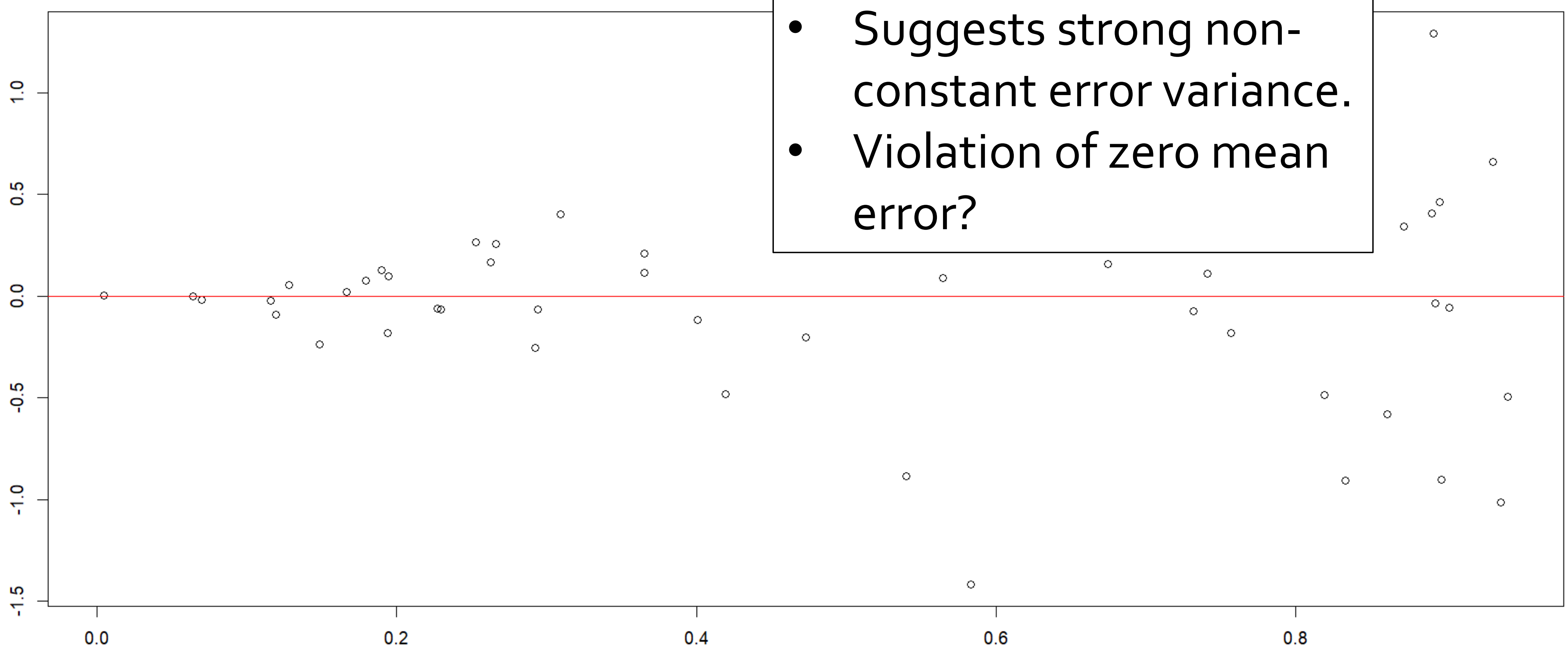
```
mean(resid(model))
```

Zero mean error + Homoscedasticity

What do you see?



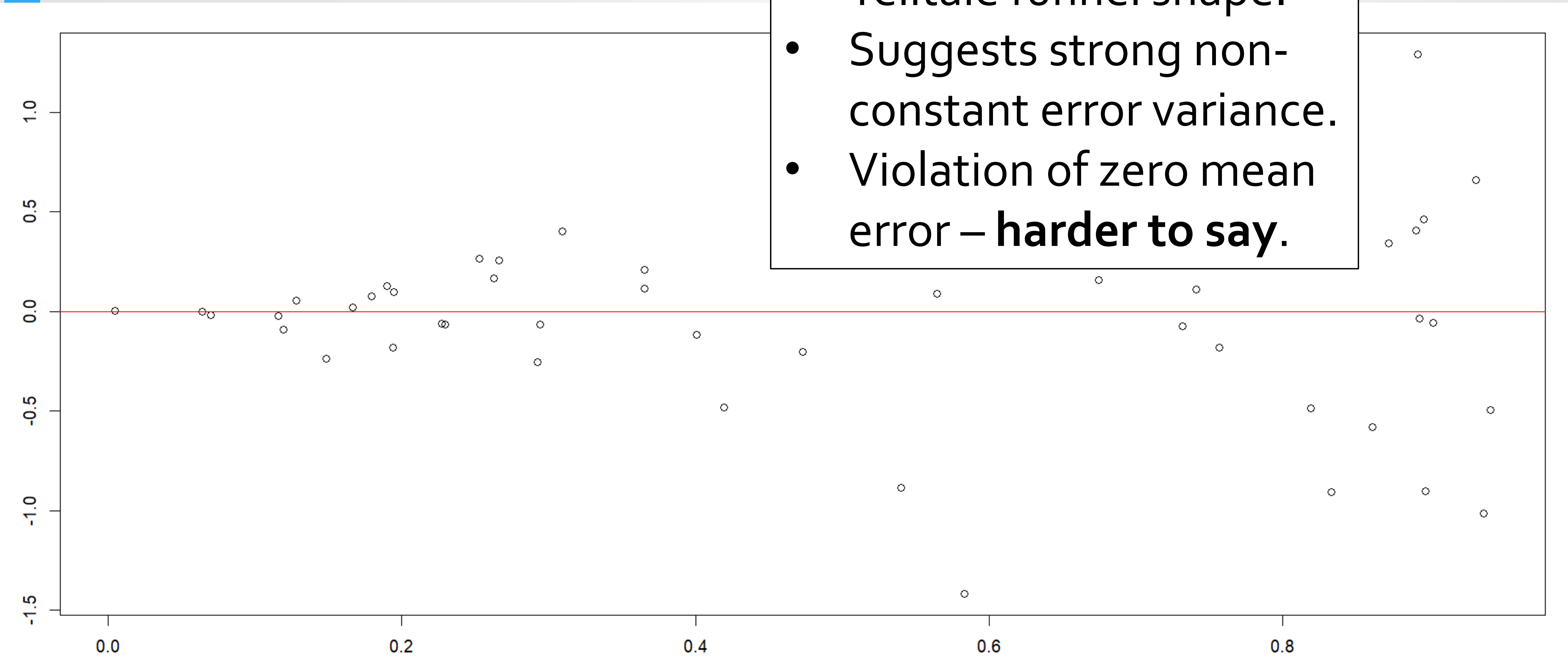
Zero mean error + Homoscedasticity



- Telltale funnel shape.
- Suggests strong non-constant error variance.
- Violation of zero mean error?

Zero mean error + homoscedasticity

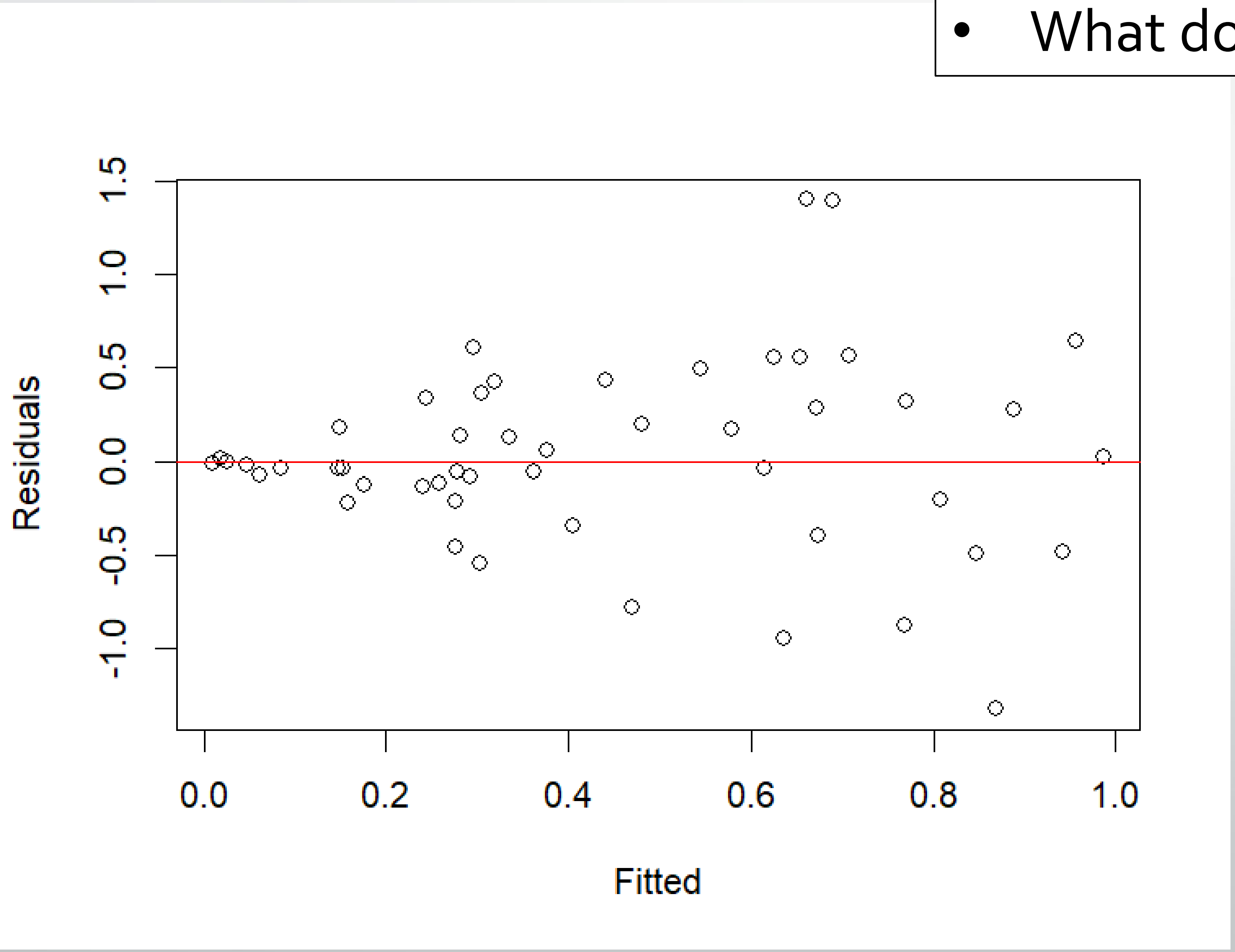
- Telltale funnel shape.
- Suggests strong non-constant error variance.
- Violation of zero mean error – **harder to say**.



Zero mean error + homoscedasticity

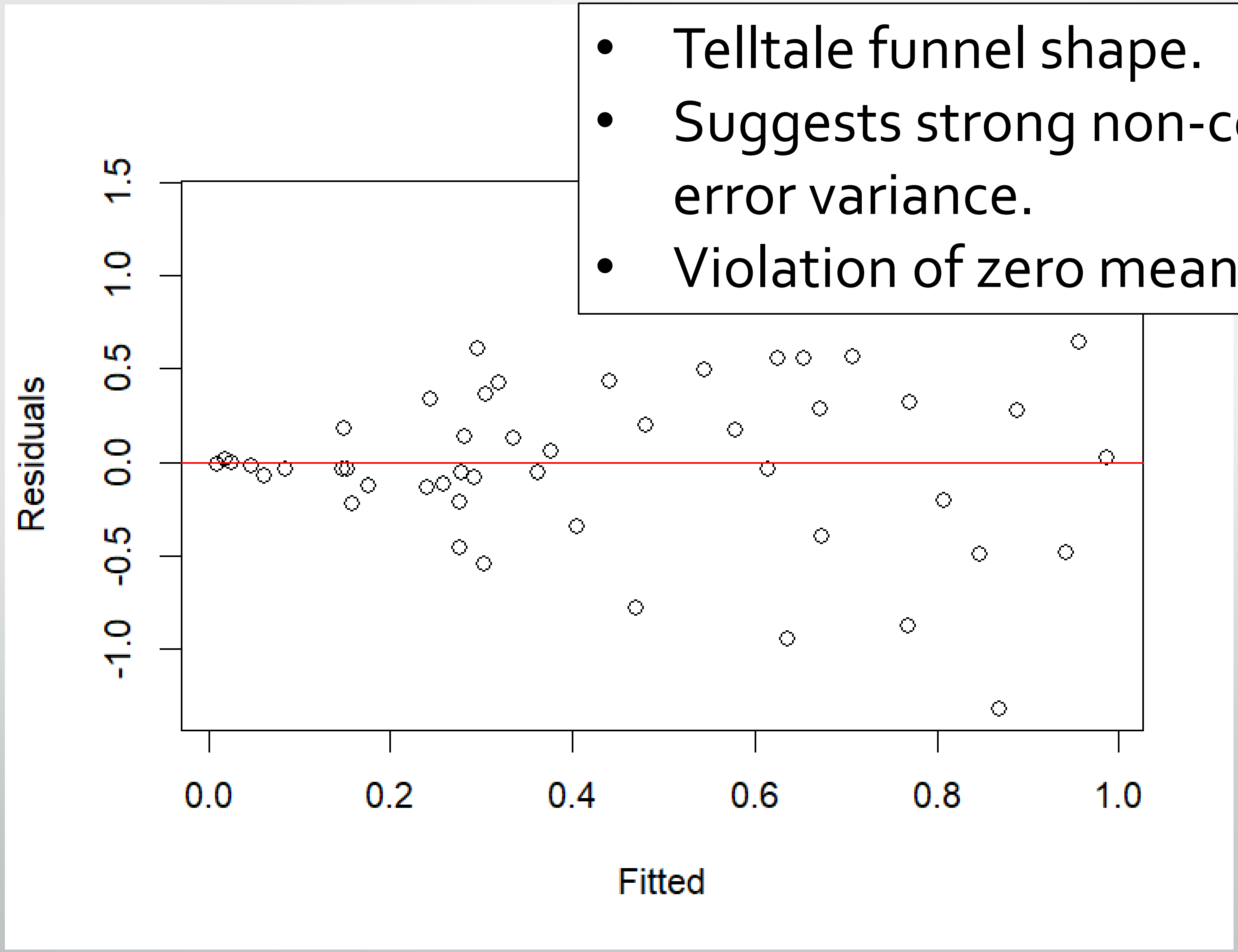
- Let’s look at another example.
- This time we’ll assess the zero mean error and homoscedasticity assumptions both graphically and numerically.

Zero mean error + homoscedasticity



- What do you see?

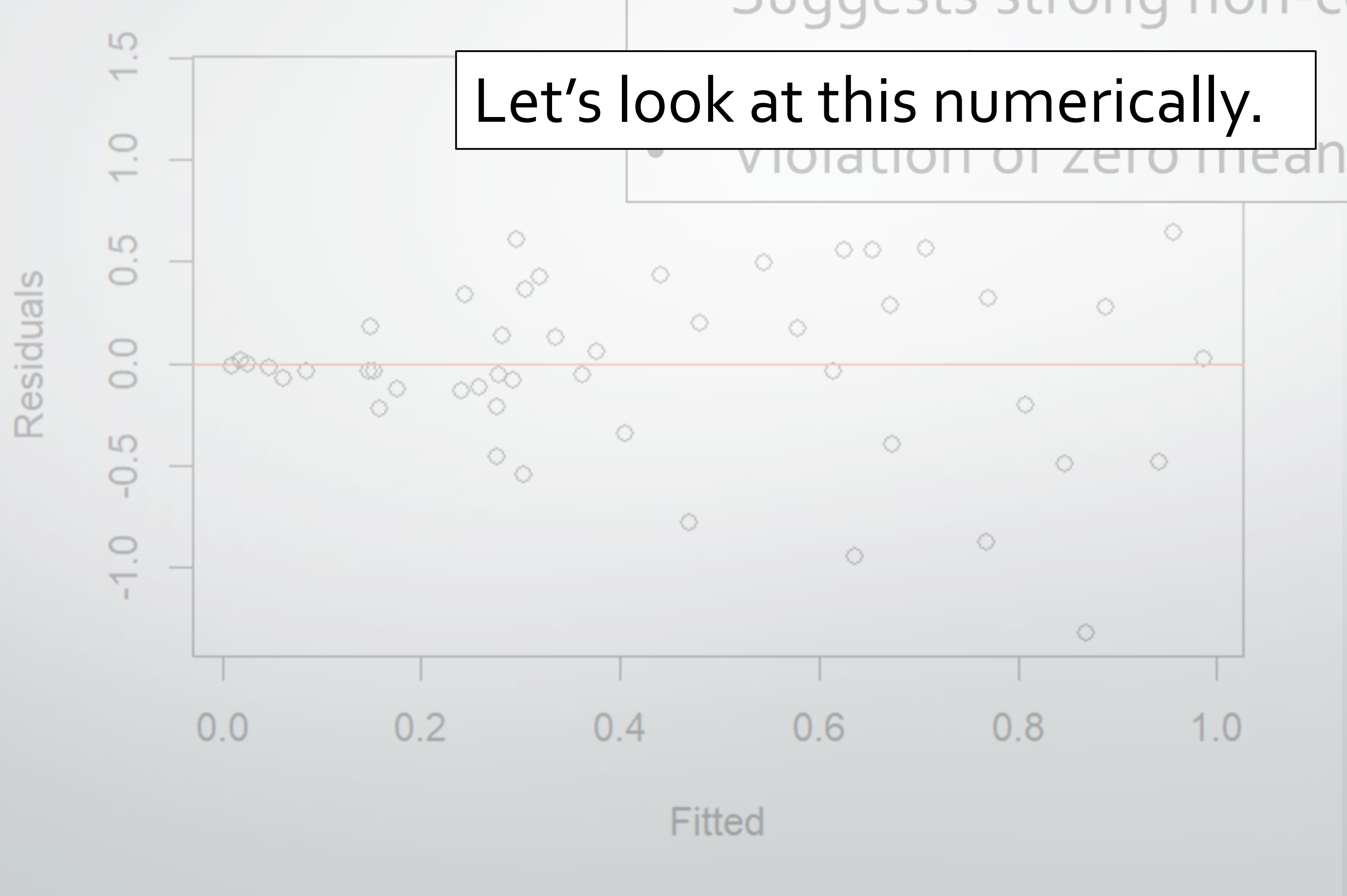
Zero mean error + homoscedasticity



- Telltale funnel shape.
- Suggests strong non-constant error variance.
- Violation of zero mean error?

Zero mean error + homoscedasticity

- Telltale funnel shape.
- Suggests strong non-constant
- Violation of zero mean error?



Let's look at this numerically.

Zero mean error - numerically

How can we assess whether the mean of our errors is 0?

- `mean(resid(model))`

What if we wanted a p-value test whether the mean is different from 0?

- `t.test(resid(model))`
- `wilcox.test(resid(model))`

Zero mean error - numerically

How can we assess whether the mean of our errors is 0?

- `mean(resid(model))`

```
> mean(resid(m1))
[1] 1.145134e-17
```

What if we wanted a p-value test whether the mean is different from 0?

- `t.test(resid(model))`
- `wilcox.test(resid(model))`

```
> t.test(resid(m1))

        One Sample t-test

data:  resid(m1)
t = 1.6177e-16, df = 49, p-value = 1
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.1422525  0.1422525
sample estimates:
   mean of x
1.145134e-17
```

```
> wilcox.test(resid(m1))

        Wilcoxon signed rank test with continuity correction

data:  resid(m1)
V = 632, p-value = 0.9615
alternative hypothesis: true location is not equal to 0
```

Homoscedasticity - numerically

How can we assess whether the variance of our errors is constant?

- Partition/subset residuals and compare using F test
  - `var.test(dat$residuals[dat$fitted <0.4], dat$residuals[dat$fitted >=0.4])`

What else might we do?

- Examine for correlation between residuals and fitted values.
  - `cor(dat$residuals, dat$fitted)`
- Fit linear model comparing residuals and fitted values.
  - `summary(lm(residuals ~ fitted, dat))`

Homoscedasticity - numerically

How can we assess whether the variance of our errors is constant?

- Partition/subset residuals and compare using F test
  - `var.test(dat$residuals[dat$fitted <0.4], dat$residu` [illegible]

What do you expect to see with each of these tests?

What else might we do?

- Examine for correlation between residuals and fitted values.
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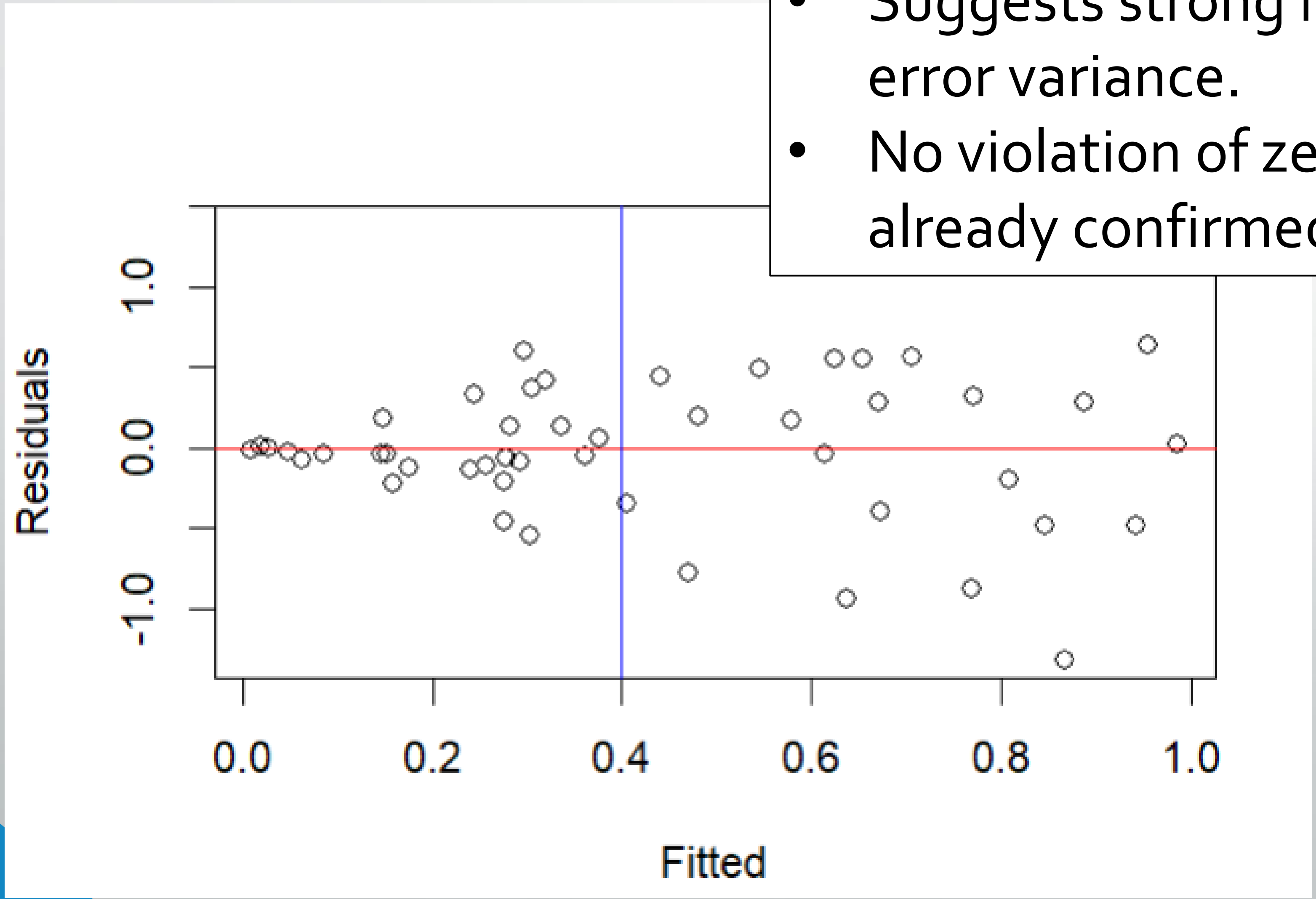
Zero mean error + homoscedasticity



- Telltale funnel shape.
- Suggests strong non-constant error variance.
- No violation of zero mean error already confirmed.

Homoscedasticity - numerically

How can we assess whether the variance of our errors is constant?

- Partition/subset residuals and compare using F test
  - `var.test(dat$residuals[dat$fitted <0.4], dat$residuals[dat$fitted >=0.4])`

```
> var.test(dat$residuals[dat$fitted <0.4], dat$residuals[dat$fitted >=0.4])

        F test to compare two variances

data:  dat$residuals[dat$fitted < 0.4] and dat$residuals[dat$fitted >= 0.4]
F = 0.13192, num df = 25, denom df = 23, p-value = 3.744e-06
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.05768073 0.29749247
sample estimates:
ratio of variances
         0.1319215
```

Homoscedasticity - numerically

How can we assess whether the variance of our errors is constant?

- Examine for correlation between residuals and fitted values.
  - `cor(dat$residuals, dat$fitted)`

```
> cor(dat$fitted, dat$residuals)
[1] 0.02424518
```

- Fit linear model comparing residuals and fitted values.
  - `summary(lm(residuals ~`

```
> summary(lm(residuals ~ fitted, dat))

Call:
lm(formula = residuals ~ fitted, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.36993 -0.21725 -0.02996  0.26965  1.35893 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.01052    0.13442   0.078    0.938
fitted       0.04313    0.25668   0.168    0.867

Residual standard error: 0.5057 on 48 degrees of freedom
Multiple R-squared:  0.0005878, Adjusted R-squared:  -0.02023 
F-statistic: 0.02823 on 1 and 48 DF,  p-value: 0.8673
```

Zero mean error + homoscedasticity

Zero mean error

- Can be difficult to assess graphically.
- Better to assess by taking mean of the residuals → can get p-value if needed.

Homoscedasticity

- Should **always** assess graphically.
- For less obvious cases can stratify errors and compare variance of strata using F test.
- Checking correlation/association between residuals and fitted values generally insufficient.

Zero mean error + homoscedasticity

Zero mea

- Can be d
- Better to ... et p-value if needed.

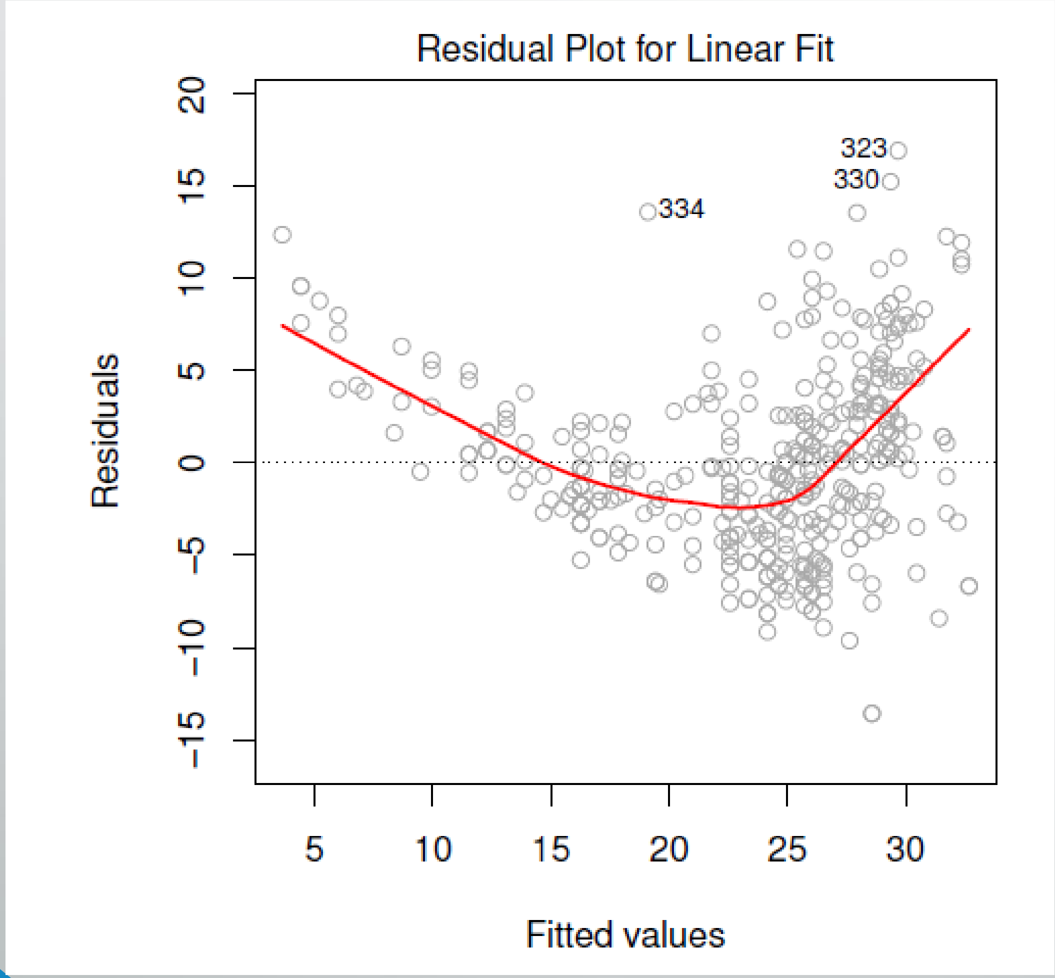
Homosce

- Should a
- For less c ... riance of strata usi
- Checking ... fitted values generally

There are a number of other tests that you may come across which aim to assess homoscedasticity (e.g., Levene, Brown-Forsythe, Breusch-Pagan, Bartlett).

****Do not use these in this class. Each has its own limitations which we will not go into, so stick with the approaches we've discussed.****

Zero mean error + error normality



ISLR Figure 3.9

Can clearly see that the residuals are not normally distributed.

How else might we assess error normality?

- Graphically?
- Statistically?

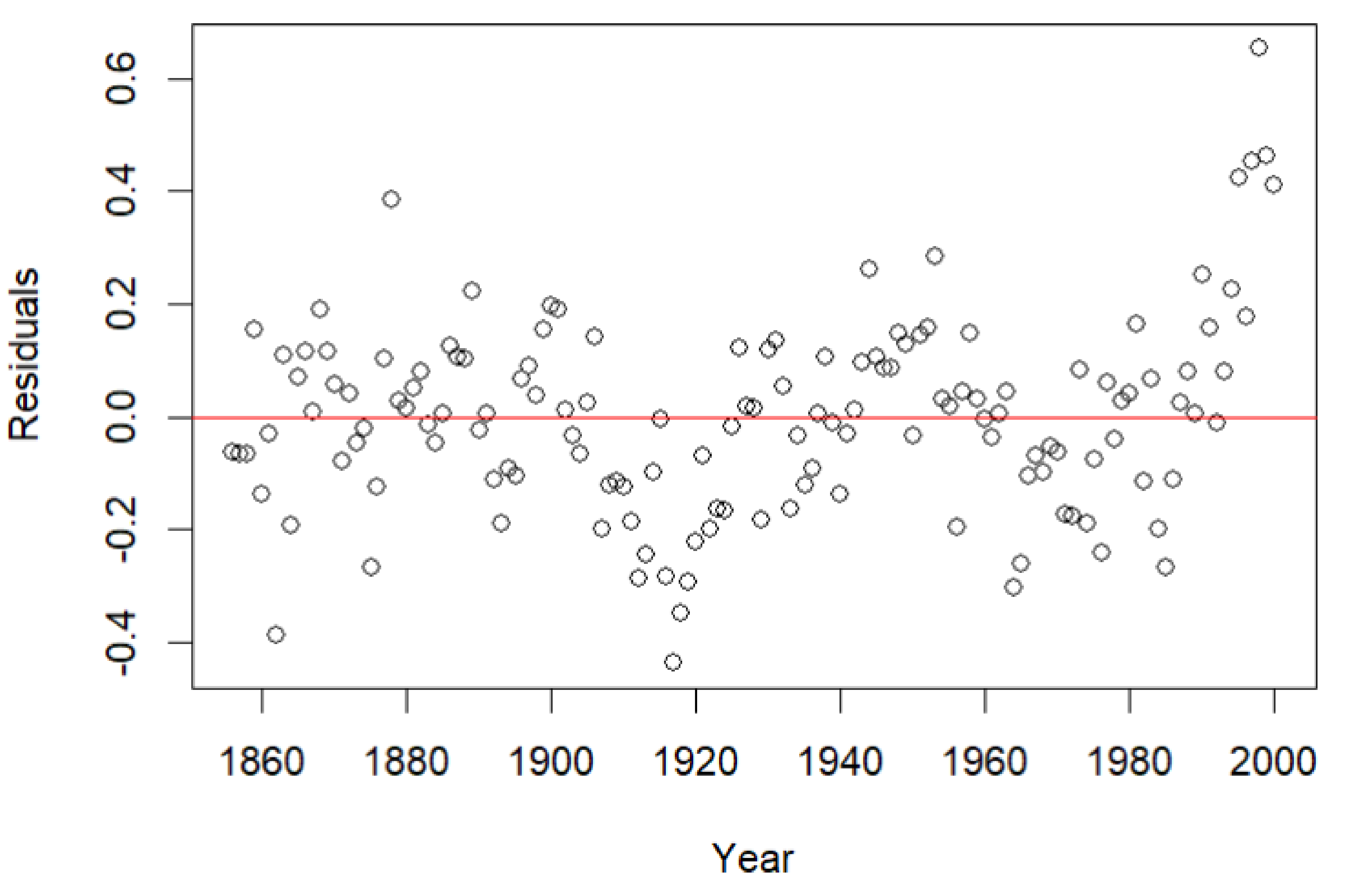
How else might we assess error normality?

- Graphically?
  - QQ plot
- Statistically?
  - Shapiro-Wilk test
  - Should not be relied on as it may suggest departures from normality when there really are none—particularly with large sample sizes.

Uncorrelated errors

- One of the most important assumptions to assess, but unfortunately also difficult to assess given the numerous patterns correlation can take.
- Some common examples when errors may be correlated:
  - Time series data: (e.g., temperature over time, cases of influenza over time, etc.)
  - Clustered data: (e.g., geographic proximity, repeated measures, etc.)
- Serial correlation like that seen in time series data can sometimes be identified using residual plots.
  - Often, the potential for correlated errors is something you need to consider when assessing the type of data you're using.

Uncorrelated errors



Unusual observations

Unusual observations

Leverage

- How similar the value of an observation is in relation to the majority of values for that variable.

Outliers

- How similar is an observation compared to what the regression estimates for that value.

Influential points

- Removal of an influential observation from the data results in a large change in the model fit.

Unusual observations

Leverage

- Measured with hat and internally studentized residuals
  - `hatvalues(model)` → should sum to number of parameters in model
    - Want to examine hat values > 2(mean(hatvalues(model)))
    - Larger hat value → greater leverage (often examined using half normal plot)
  - `rstandard(model)`

Outliers

- Measured through externally studentized residuals – `rstudent()`
  - Can assess whether largest residuals are greater than the critical value by:
  - `sort(abs(r_ex),dec=T)[1:10]>abs(qt(.05/(10),120-2))`

Influential points

- Measured with Cook's Distance – `cooks.distance(model)`

Unusual observations

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Influential points

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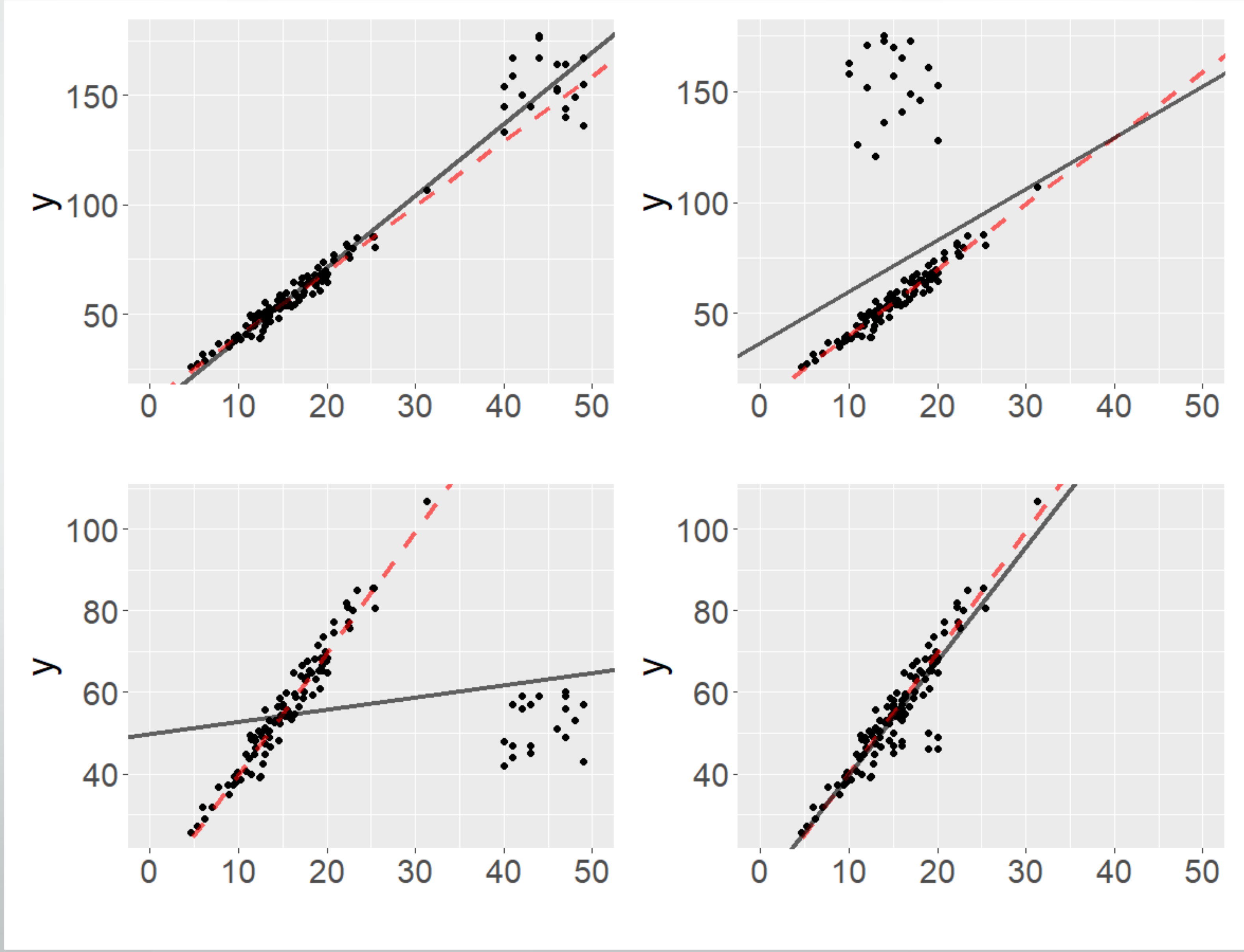
Perfect_add2-perfect_add5 Models

What do you points do you think have high leverage?

Are outliers?

Are influential?



Multiple testing

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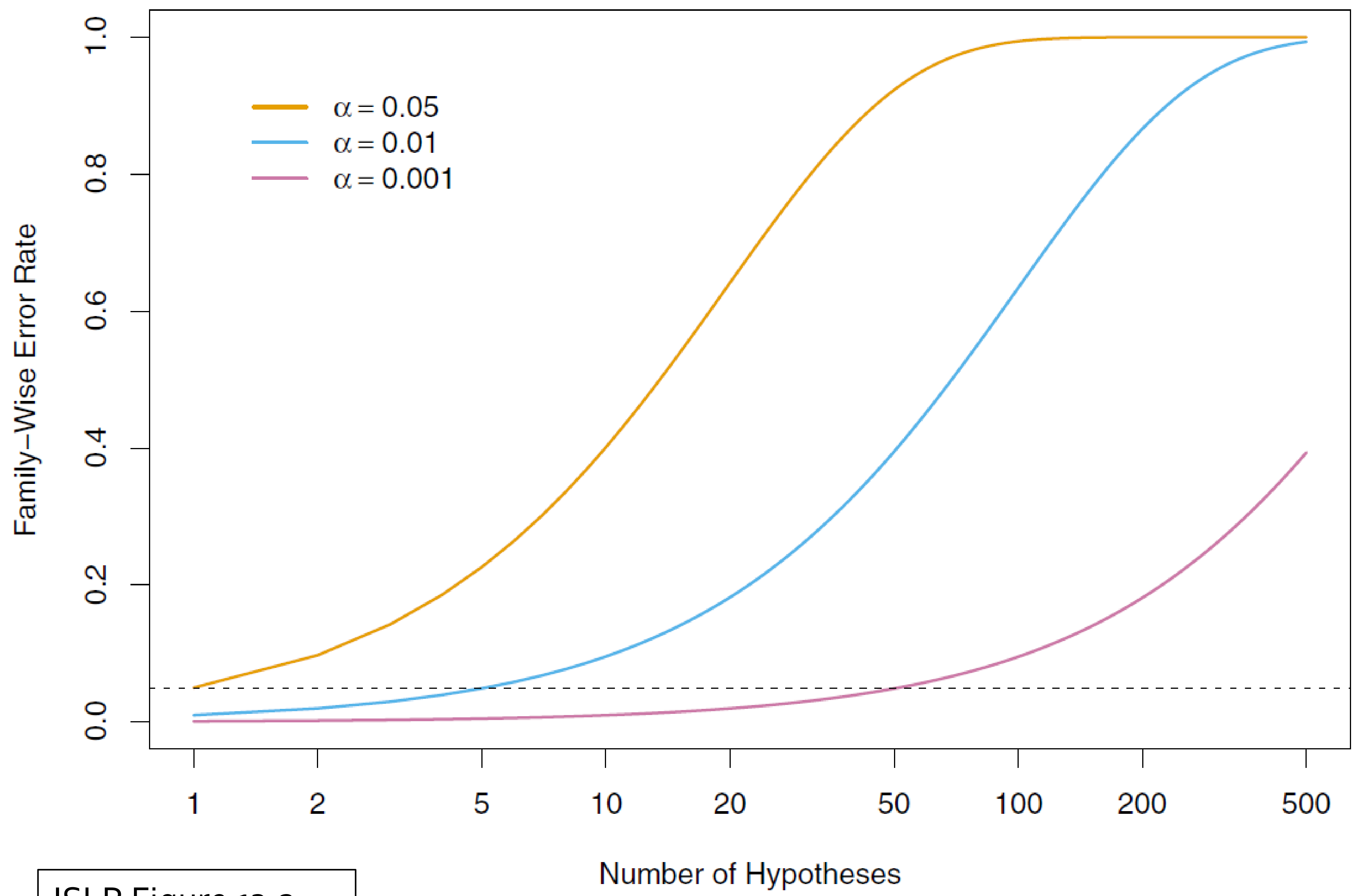
- We're making 10 comparisons, so under the Bonferroni correction we divide alpha by 10.

What does this do?

Multiple testing



ISLR Figure 13.2